

Conduct and Testing For Conduct

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Grad IO

The problem

Conduct Testing in Industrial Organization

Foundational Empirical IO Question: How do we observe data on price and quantity and infer which model of firm behavior generated those outcomes?

- ▶ Early work: Porter (1983), Bresnahan (1982,1987)
- ▶ Subsequent work defined the “menu” approach: Nevo (1998, 2001), Villas-Boas (2007)
- ▶ Recent revival of “internalization” parameters: Miller and Weinberg (2017), Crawford, Lee, Whinston, and Yurukoglu (2017), Pakes (2017)
- ▶ Parallel work by: Duarte, Magnolfi, Sølvesten, Sullivan (QE 2024): which test is best (RV) and when instruments are too weak to test. Magnolfi, Quint, Sullivan, Waldfogel (2022): should we test or estimate?
- ▶ Applications of our test: Starc and Wollmann (2022), Scuderi and Roussille (2025), others?
- ▶ Today: the classic approaches, then the Backus–Conlon–Sinkinson scalar test (“BCS 1.0”) and the full conditional-moment test (“BCS 2.0”), then what the cereal data taught us about what identifies conduct.

Is conduct testable? Berry and Haile (2014): yes.

- ▶ Absent additional restrictions, we cannot generally look at data on (P, Q) and decide whether or not collusion is taking place.
 - You say we started colluding at date t , I say we received a correlated shock to mc .
- ▶ We can make progress in two ways: (1) parametric restrictions on marginal costs; (2) exclusion restrictions on supply.
 - Most of the literature focuses on (1) by assuming something like: $\ln mc_{jt} = x_{jt}\gamma_1 + w_{jt}\gamma_2 + \omega_{jt}$.
 - In principle (2) is possible if we have instruments that shift demand for products but not supply. (These are much easier to come up with than “supply shifters”). (This is the point Berry Haile 2014 make).

A famous plot (Bresnahan 87)

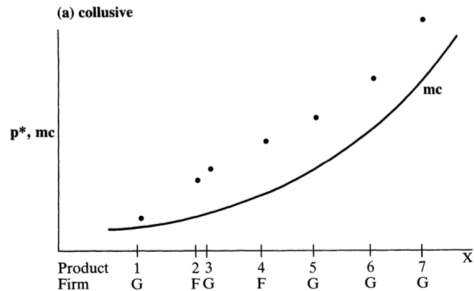


Figure 2(a)

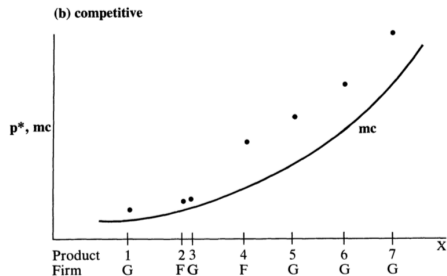


Figure 2(b)

Bresnahan (1980/1982) recognized this problem: we need “rotations of demand”.

Conduct Testing in Pictures (Berry Haile 2014)

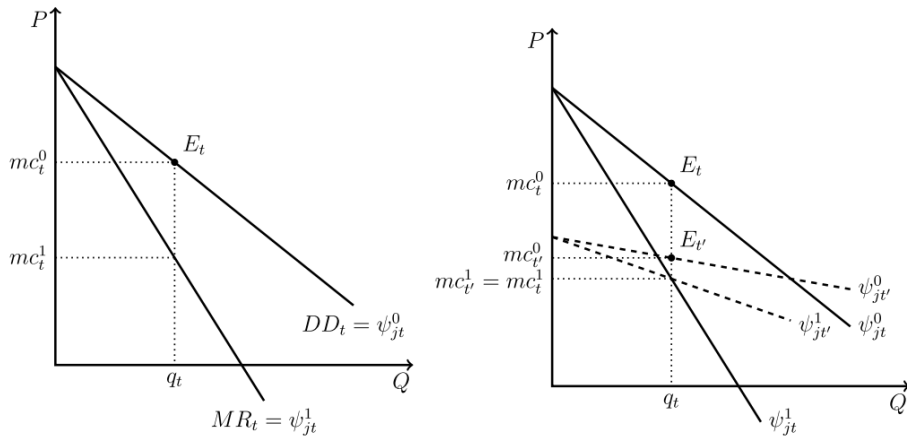


Figure 2(ab) from Berry and Haile (2014), Example 1.

We begin with a relatively standard BLP-style differentiated products setup.

- ▶ Markets t
- ▶ Products j
- ▶ Data $\chi_t = \{(x_{jt}, v_{jt}, w_{jt}) \text{ for all } j \in \mathcal{J}_t\}$.
- ▶ Market Shares $\mathbf{s}_t = [s_{1t}, \dots, s_{Jt}, s_{0t}]$.
- ▶ Prices $\mathbf{p}_t = [p_{1t}, \dots, p_{Jt}]$.
- ▶ Consumers i with demographics y_{it} (income, presence of kids, etc.)

Testing Conduct: Multiproduct Bertrand Example

We generalize the $\mathcal{H}(\kappa)$ and derive multi-product Bertrand FOCs:

$$\begin{aligned}\arg \max_{p \in \mathcal{J}_f} \pi_f(\mathbf{p}) &= \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p}) + \kappa_{fg} \sum_{j \in \mathcal{J}_g} (p_j - c_j) \cdot q_j(\mathbf{p}) \\ \rightarrow 0 &= q_j(\mathbf{p}) + \sum_{k \in (\mathcal{J}_f, \mathcal{J}_g)} \kappa_{fg} \cdot (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})\end{aligned}$$

- Instead of 0's and 1's we now have $\kappa_{fg} \in [0, 1]$ representing how much firm f cares about the profits of g .
- If f and g merge (or fully coordinated) then $\kappa_{fg} = 1$
 - Often in the real world firms cannot reach fully collusive profits and $\kappa_{fg} \in (0, 1)$.
 - Evidence that $\kappa_{fg} > 0$ is not necessarily evidence of malfeasance, just a deviation from **static Bertrand pricing** (e.g. Cournot, Dynamics, etc.)

Testing Conduct: Multiproduct Bertrand Example

- ▶ Recall the Δ matrix which we can write as $\Delta = \tilde{\Delta} \odot \mathcal{H}(\kappa)$, where $\odot$ is the element-wise or Hadamard product of two matrices.
 - $\tilde{\Delta}$ is the matrix of demand derivatives with $\Delta(j, k) = \frac{\partial q_j}{\partial p_k}$ for all elements.
 - $\mathcal{H}(\kappa) = \kappa_{fg}$ for products owned by (f, g) where $\kappa_{ff} = 1$ always.
- ▶ Mergers are about changing 0's to 1's in the $\mathcal{H}(\kappa)$ matrix.
- ▶ Matrix form of FOC: $q(\mathbf{p}) = \Delta(\mathbf{p}, \kappa) \cdot (\mathbf{p} - \mathbf{mc})$
- ▶ $\mathbf{mc} = \mathbf{p} - \underbrace{\Delta(\mathbf{p}, \theta_2, \kappa)^{-1} \boldsymbol{\sigma}(\mathbf{p})}_{\eta(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)}$ where η_{jt} is the markup.

Reasons for Deviations from Multiproduct Bertrand

Biased estimates of own and cross price derivatives: For anything to work, you have correct estimates of $\tilde{\Delta}$. My prior is most papers **underestimate** diversion ratios for close substitutes.

Vertical Relationships: Who sets supermarket prices? Just the retailer? Just the manufacturer? Some combination of both? Retailers tend to **soften** downstream price competition.

Faulty Timing Assumptions: Bertrand is a simultaneous move pricing game. Lots of alternatives (Stackelberg leader-follower, Edgeworth cycles, etc.).

Dynamics and Dynamic Pricing: Forward looking firms or consumers might not set static Nash prices. [e.g. Temporary Sales, Switching Costs, Network Effects, etc.]

Unmodeled Supergame: Maybe firms are legally tacitly colluding, higher prices might be about what firms believe will happen in a price war.

Assume additivity, and write in terms of structural errors:

$$\begin{aligned}\delta_{jt}(\mathbf{s}_t, \mathbf{y}_t, \tilde{\theta}_2) + \alpha p_{jt} &= h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}, \theta_1) + \xi_{jt} \\ f(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)) &= h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}\end{aligned}$$

- ▶ To simplify slides we let $f(x) = x$ (often $f(x) = \log(x)$) but we can put that in $h_s(\cdot)$.
- ▶ $h(\cdot)$ are often just linear relationships like: $\theta_1[\mathbf{x}_{jt}, \mathbf{v}_{jt}]$.
- ▶ Endogeneity Problem: p_{jt} and η_{jt} are functions of (ξ, ω) .
- ▶ (θ_2, κ) parameters that determine markups

Two Ways In (Recap of the BLP Lecture)

1. **Demand first.** Estimate θ_2 from demand alone with $\mathbb{E}[\xi_{jt} \mid \mathbf{x}_t, \mathbf{v}_t, \mathbf{w}_t] = 0$; recover $\widehat{\mathbf{mc}} = \mathbf{p} - \boldsymbol{\eta}(\mathbf{p}, \mathbf{s}, \hat{\theta}_2, \kappa)$ for **any** κ .
2. **Jointly.** Stack the supply moments $\mathbb{E}[\omega_{jt} \mid \mathbf{x}_t, \mathbf{w}_t, \mathbf{v}_t] = 0$ with demand; the supply side sharpens α (BLP 1995) but imports whatever conduct you assumed.

The problem with either: given $[\mathbf{q}, \mathbf{p}, \widetilde{\Delta}, \mathcal{H}(\kappa)]$ there is **always** a vector $\mathbf{mc}$ that rationalizes the data (J equations, J unknowns). Nonparametrically κ is not identified without more.

- ▶ Restriction 1: a model of marginal cost, $f(p_{jt} - \eta_{jt}(\kappa)) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}; \theta_3) + \omega_{jt}$, so that “ $mc < 0$ ” or “ mc jumps around” become evidence.
- ▶ Restriction 2: **exclusion**. Something that moves markups but not costs, $\mathbb{E}[\omega_{jt} \mid \mathbf{z}_{jt}] = 0$ (Berry Haile 2014: demand rotators $\mathbf{v}_t$, rivals’ characteristics).
- ▶ Everything that follows is a way of turning those two restrictions into a test. The second one does all the real work.

Classic approaches

There are 2.5 ways to think about conduct:

1. Using moment conditions to estimate $\hat{\kappa}$ or $\mathcal{H}(\kappa)$ directly. (Wald)
 - Often with a small number of parameters (ie: $\kappa_{fg} = 0$ except for firms I know are in a cartel).
 - Can be challenging to tell similar values of κ_{fg} apart (under-powered).
2. “Menu Approach” / (LR)
 - Nevo (Economics Letters 1998)
 - Bresnahan (1987)
 - Compare some goodness-of-fit criteria across assumed values of κ (Bertrand vs. Collusion)
3. Rejecting (or failing to reject) a single model.
 - “We can reject monopoly pricing...”

Put the η_{jt} on the RHS and test whether $\lambda = 1$:

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \lambda \cdot \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa) + \omega_{jt} \text{ with } \mathbb{E}[\omega_{jt} | \mathbf{x}_t, \mathbf{w}_t, z_t] = 0$$

- ▶ We are basically running 2SLS with IV for the endogenous η_{jt}
- ▶ “Informal” test of Villas Boas (2007): $\mathbb{E}[\omega_{jt} | \mathbf{x}_{jt}, \mathbf{w}_{jt}, \eta_{jt}] = 0$.
 - Considers different forms of $f(\cdot)$: linear, exponential, logarithmic.
 - WP discusses this more than Restud version.
- ▶ Pakes (2017) uses Wollman (2018) data and BLP IV $\mathbb{E}[\omega_{jt} | \mathbf{x}_{jt}, \mathbf{w}_{jt}, f(\mathbf{x}_{-j})] = 0$.
- ▶ $\lambda \neq 0$ is hard to interpret.

Table 1: Wollman & Pricing Equilibrium.

Taken from Pakes, 2017, *Journal of Industrial Economics*.

	Price (S.E.)		Price (S.E.)	
Gross Weight	.36	(0.01)	.36	(.003)
Cab-over	.13	(0.01)	.13	(0.01)
Compact front	-.19	(0.04)	0.21	(0.03)
long cab	-.01	(0.04)	0.03	(0.03)
Wage	.08	(.003)	0.08	(.003)
<i>Markup</i>	.92	(0.31)	1.12	(0.22)
Time dummies?	No	n.r.	Yes	n.r.
R ²	0.86	n.r.	0.94	n.r.

Note. There are 1,777 observations; 16 firms over the period 1992-2012. S.E.=Standard error.

These are somewhat reassuring:

- ▶ $\lambda \approx 1$ for multiproduct-oligopoly
- ▶ Fit is pretty good $R^2 > 0.8$ and $R^2 > 0.5$ for within vehicle regressions (not shown).
- ▶ As a behavioral model, multiproduct demand estimation seems successful.
- ▶ But, do we know that an alternative $\mathcal{H}(\kappa)$ would have a $\lambda \neq 1$ or a lower R^2 , and if so how low before we can “reject” the model?

Another idea (Bonnet and Dubois, Rand 2010) runs the following regression:

$$\log(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \hat{\theta}_2, \kappa)) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}$$

- ▶ Run a regression for each κ and obtain $Q(\kappa) = \sum_{jt} \hat{\omega}_{jt}^2$
- ▶ Employ the **non nested test** of Rivers and Vuong (2002). Why?
- ▶ Working out the distribution of $Q(\kappa_1) - Q(\kappa_2) = T(\kappa_1, \kappa_2)$ is the hard part.
- ▶ Also this is OLS (or NLLS) and there are no instruments or **exclusion restrictions** for the supply side. Presumably we could add some and do GMM? (I think this is the “formal” test of Villas Boas (ReStud 2007)).

TABLE 7

Percentage price-cost margins (PCM) by scenario

Wholesale, retail, and total PCM	Median	S.D.	Min	Max
Given random coefficient demand				
Model 1: Simple linear pricing: T-PCM	43.8	25.4	25.5	136.5
Model 2: Hybrid model: T-PCM	41.6	28.6	15.8	136.7
Model 3.1: Zero wholesale margin: R-PCM	21.1	9.0	12.5	50.0
Model 3.2: Zero retail margin: W-PCM	20.6	4.7	12.8	45.8
Model 4: Wholesale collusion: T-PCM	72.8	22.4	38.8	253.0
Model 5: Retail collusion: T-PCM	69.9	25.8	23.4	263.2
Model 6: Monopolist: T-PCM	40.6	11.5	21.8	103.7

PCM = $(p - c)/p$ where p is price and c is marginal cost. S.D., standard deviation; W, wholesale; R, retail; T, total.

Source: My calculations.

TABLE 8

Sample statistics of recovered costs and informal testing

	Recovered retail and wholesale costs				
	Mean	S.D.	Min	Max	Percentage <0
Model 1: Simple linear pricing	0.2730	0.1552	-0.1532	0.5364	1.35
Model 2: Hybrid model	0.2757	0.1681	-0.1541	0.5364	1.45
Model 3.1: Zero wholesale margin	0.3796	0.0907	0.1201	0.6298	0
Model 3.2: Zero retail margin	0.3815	0.0895	0.1302	0.6279	0
Model 4: Wholesale collusion	0.1352	0.1114	-0.4590	0.4409	13.12
Model 5: Retail collusion	0.1530	0.1329	-0.4895	0.5053	12.6
Model 6: Monopolist	0.2953	0.0951	-0.0089	0.5628	0.08

Notes: Recovered costs = $p - \text{epcm}$ where p is retail price and epcm are the estimated margins. Last column has the percentage of cases with recovered estimated costs being negative.

Source: My calculations.

- Try out different models of price setting behavior, compute η markups.
- Unsurprisingly models with higher markups also have more costs $MC < 0$.
- Is this evidence of incorrect conduct assumption or inelastic demand?

TABLE 10

p-Values for pairwise non-nested comparisons

H_0 model	Alternative models						
	1	2	3-1	3-2	4	5	6
1: Simple linear pricing	–	0.50	0.00	0.50	0.24	0.00	0.50
2: Hybrid	0.00	–	0.50	0.50	0.12	0.00	0.50
3.1: Zero wholesale margin	0.41	0.29	–	0.05	0.50	0.39	0.07
3.2: Zero retail margin	0.39	0.40	0.05	–	0.50	0.39	0.17
4: Wholesale collusion	0.49	0.48	0.50	0.50	–	0.48	0.50
5: Retail collusion	0.00	0.00	0.50	0.50	0.22	–	0.50
6: Monopolist	0.34	0.35	0.17	0.31	0.48	0.34	–
<i>Chain size weighted</i>							
1: Simple linear pricing	–	0.08	0.01	0.06	0.08	0.00	0.00
2: Hybrid	0.17	–	0.15	0.22	0.00	0.06	0.14
3.1: Zero wholesale margin	0.08	0.15	–	0.11	0.15	0.12	0.00
3.2: Zero retail margin	0.01	0.07	0.00	–	0.09	0.01	0.00
4: Wholesale collusion	0.00	0.05	0.04	0.09	–	0.00	0.02
5: Retail collusion	0.00	0.02	0.03	0.11	0.02	–	0.00
6: Monopolist	0.10	0.20	0.00	0.15	0.20	0.14	–

► Cox test has pairwise rejection problem
(4) rejects (5) and (5) rejects (4).

► Likewise for 3.1 and 3.2 in top panel.

Notes: *p*-Values reported from non-nested, Cox-type (Smith, 1992) test statistics of the null model in a row being true against the specified alternative model in a column. Bottom part is a robustness check. It has the same format as above, but the non-nested comparisons are based on estimates for the case when the portion of the manufacturer's profit due to each retailer is weighted by the retailer's chain size.

Source: My calculations.

So far three approaches to exploit $\mathbb{E}[\omega_{jt} | \mathbf{x}_t, \mathbf{w}_t, z_t] = 0$

1. Put the markup on RHS and instrument for it to test $\lambda = 1$ (Wald)

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \lambda \cdot \eta_{jt}(\mathbf{p}, \mathbf{s}, \widehat{\theta}_2, \kappa) + \omega_{jt}$$

2. Put the markup on LHS assuming $\lambda = 1$ and test goodness of fit of supply equation (Anderson Rubin)

$$p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \widehat{\theta}_2, \kappa) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}$$

3. Estimate supply and demand simultaneously $[\theta_1, \theta_2, \theta_3]$ and compare goodness of fit for different κ . (Likelihood Ratio)

Which Test? Duarte, Magnolfi, Sølvesten, Sullivan (QE 2024)

Take the supply equation seriously as a **moment condition** and ask which classical test to run on a menu of markups $\{\eta^m\}$:

- ▶ **Specification tests** of one model at a time (Wald on $\lambda = 1$, Anderson–Rubin / GMM- J on $\mathbb{E}[\omega^m z] = 0$, Cox): with enough data **every** model is rejected, because no conduct model is exactly right. Villas-Boas's Cox test rejecting in both directions is the symptom.
- ▶ **Model selection** (Rivers–Vuong 2002): compare two models' GMM criteria, $Q_W(\eta^{m_1}) - Q_W(\eta^{m_2})$, under the null that they are **equally** misspecified. Only RV behaves sensibly when both models are wrong, which is always.
- ▶ The catch: RV is **degenerate** when the two models are indistinguishable through the instruments. DMSS define **weak instruments for testing**, extend Stock–Yogo to model selection, and give an effective- F diagnostic with critical values for worst-case size and best-case power (pyRVtest).
- ▶ Their criterion is a finite-dimensional 2SLS quadratic form, $\bar{\mathbf{m}}'_m (z' z)^{-1} \bar{\mathbf{m}}_m$ with $\bar{\mathbf{m}}_m = \frac{1}{n} \sum_i z_i \omega_i^m$ (a K -vector of unconditional moments). Keep that in mind: the two BCS criteria below are a **single optimal direction** and the **full conditional-moment norm**, and the DMSS object is the finite- K thing in between.

Model Selection under Misspecification

What Is the Null? Ranking Misspecified Models

Each conduct m implies markups η_{jt}^m , hence an implied cost and a cost residual:

$$c_{jt}^m = p_{jt} - \eta_{jt}^m = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^m, \quad \mu_m(z) \equiv \mathbb{E}[\omega_{jt}^m \mid z_{jt}], \quad Q(\eta^m) \equiv \|\mu_m\|^2 = \mathbb{E}[\mu_m(z)^2] \geq 0.$$

- ▶ Correct conduct: $\mu_m \equiv 0$ and $Q(\eta^m) = 0$. On real data **every** $Q(\eta^m) > 0$: no model on the menu satisfies the restriction exactly.
- ▶ So the honest hypothesis is not “Bertrand is true” but Vuong (1989) / Rivers–Vuong (2002) / Chen–Hong–Shum (2007):

$$H_0 : Q(\eta^{m_1}) = Q(\eta^{m_2}) \quad (\text{equally wrong}) \quad \text{vs.} \quad H_A : \text{one is closer to the moment condition.}$$

- ▶ A “test of conduct” is therefore a **ranking**: which markups are closest to $\mathbb{E}[\omega \mid z] = 0$. “Model m is in the model confidence set” means it is the hardest to beat, not that it is correct.
- ▶ The cost function h_s is a nuisance; it is the **pseudo-true** cost function under each (possibly wrong) conduct, well defined because the criterion is shared across models.

Degeneracy: When Two Wrong Models Are the Same Wrong Model

- ▶ The RV statistic is $\sqrt{n}(\widehat{Q}_1 - \widehat{Q}_2)/\widehat{\sigma}$. Its denominator is the variance of the **difference** of the two models' moments, and it can be zero under the null.
- ▶ That happens when the two conducts induce the **same conditional moment**, $\mu_{m_1}(z) \equiv \mu_{m_2}(z)$: observational equivalence (Chen–Hong–Shum's H_E). No instrument can rank them; the normal approximation is meaningless; $\widehat{\sigma} \rightarrow 0$ and the t -statistic is $0/0$.
- ▶ Three roads lead there:
 1. **Weak instruments**: z does not move the markup difference $\Delta\eta = \eta^{m_1} - \eta^{m_2}$ (DMSS).
 2. **Nearly identical markups**: in cereal, $\text{corr}(\eta^{m_1}, \eta^{m_2}) \geq 0.92$ across the whole menu, so $\Delta\eta$ is tiny relative to η .
 3. **Both correct** (the textbook case, and the least relevant one).
- ▶ The empirically relevant comparisons sit **near** this boundary, not on it and not far from it. Everything in the rest of the lecture is about doing inference there.

A Different Degeneracy: Constant-Markup Conducts Are a Straw Man

Suppose conduct m implies a **constant proportional markup**, $p = \phi \cdot mc$ (perfect competition is $\phi = 1$; symmetric CES-type pricing is $\phi > 1$). Then

$$\omega_\phi = p/\phi - h_s(\mathbf{x}, \mathbf{w}) = \frac{1}{\phi} \varrho, \quad \varrho \equiv \text{the price residual on } (\mathbf{x}, \mathbf{w}), \quad Q(\phi) = \frac{1}{\phi^2} \|\mathbb{E}[\varrho \mid \mathbf{z}]\|^2.$$

- ▶ The moments are **proportional**, $\mu_\phi = \mathbb{E}[\varrho \mid \mathbf{z}]/\phi$, so $\Delta \tilde{Q} = \|\mathbb{E}[\varrho \mid \mathbf{z}]\|^2(\phi_1^{-2} - \phi_2^{-2})$: τ_{RV} is normal with power and consistently selects the **larger** markup. Scale, not fit; no instrument fixes it (lossless).
- ▶ So “we reject perfect competition” means only that the price residual projects onto $\mathbf{z}$. It is a statement about demand-side variation in prices, not about conduct.
- ▶ Rules: never compare two constant-markup conducts; treat perfect competition as a sanity check, not an informative comparison. Repairs: outside information on the markup *level* (Nevo 2001), or studentizing each moment, which leaves nothing to test.
- ▶ The informative comparisons are between conducts that imply **different patterns** of markups across products and markets: single-product vs. own-firm vs. common ownership vs. monopoly. Those differ in how much of the cross-price structure the firm internalizes.

Backus, Conlon, Sinkinson: the scalar test (BCS 1.0)

We start with marginal revenue and marginal cost (unobserved ω , observed $h_s(\cdot)$)

$$\begin{aligned}mr_{jt}^m &= mc_{jt} \\ p_{jt} - \eta_{jt}^m &= h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^m\end{aligned}$$

- ▶ Let's be vague/flexible with $h_s(\cdot)$ for now, but I don't know the production function.
- ▶ Assume: Demand and hence η_{jt}^m are **known (given conduct)**.
- ▶ Idea (η^{m_1}, η^{m_2}) are own-profit vs. common ownership, or single-product vs. own-profit, or Cournot vs. Bertrand.

The Question

Two competing markups ($\eta_{jt}^{m_1}, \eta_{jt}^{m_2}$): which fits the data better?
(both may be misspecified)

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \tau \eta_{jt}^{m_1} + (1 - \tau) \eta_{jt}^{m_2} + \omega_{jt}$$

Model is defined by a conditional moment restriction $\mathbb{E}[\omega_{jt} | z_{jt}] = 0$

- ▶ $\tau = 1$ is m_1 , $\tau = 0$ is m_2 . The null is **equal fit**, $\tilde{Q}(\eta^{m_1}) = \tilde{Q}(\eta^{m_2})$, not “ $\tau = 1$ ”: both may be wrong.
- ▶ This is a **model selection** problem or a **non nested testing** problem.
 - We might want to compare more than two alternatives (too bad).
- ▶ Obvious endogeneity problem with η_{jt} !

Our Idea: Motivation #2 (Misspecification)

Index the **true** model by m_0 and a wrong one by m_1 . Then,

$$p_{jt} - \eta_{jt}^{m_0} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^{m_0}.$$

To motivate a useful test, we ask what happens when we estimate supply with the **wrong** conduct model m_1 :

$$p_{jt} - \eta_{jt}^{m_1} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \underbrace{\underbrace{\eta_{jt}^{m_0} - \eta_{jt}^{m_1}}_{\equiv \Delta \eta_{jt}} + \omega_{jt}^{m_0}}_{\omega_{jt}^{m_1}}.$$

- ▶ Misspecifying conduct introduces an omitted variable: the difference in markups.
- ▶ Our test is premised on detection of this omitted variable.

The Instrument: Optimal Among Single Directions (BCS 1.0)

Collapse the moment against one direction $b(z)$: $\bar{m}_m(b) = \mathbb{E}[\omega^m b(z)]$ and $\tilde{Q}(\eta^m) = \bar{m}_m^2$. Which b discriminates m_1 from m_2 best?

$$\omega^{m_1} - \omega^{m_2} = -(\Delta\eta - \mathbb{E}[\Delta\eta \mid x, w]) \Rightarrow g(z) \equiv \mathbb{E}[\Delta\eta \mid x, w, z] - \mathbb{E}[\Delta\eta \mid x, w]$$

- ▶ g is the projection of the markup difference onto the **excluded** instruments after the cost controls are partialled out (Frisch–Waugh–Lovell / Robinson). It is the Chamberlain optimal instrument for the omitted variable $\Delta\eta$ of the previous slide.
- ▶ **Optimality**: among all single directions, g uniquely maximizes the signal-to-noise ratio for telling the two conducts apart (Cauchy–Schwarz; heteroskedasticity adds a $1/\sigma^2(z)$ weight), attaining the full conditional-moment distance $\|\mu_{m_1} - \mu_{m_2}\|$. RV on it is **lossless** in population: with $\|g\| = 1$, $\bar{m}_1^2 - \bar{m}_2^2 = \|\mu_{m_1}\|^2 - \|\mu_{m_2}\|^2$, the full-CMR ranking and null.
- ▶ **Failure set shrinks**: a generic b also fails when $b \perp \mu_{m_1} - \mu_{m_2}$ (spurious equal fit) or $b \perp \text{span}\{\mu_{m_1}, \mu_{m_2}\}$ (Vuong degeneracy); along g both collapse to observational equivalence.
- ▶ **The subtraction is not redundant** even though $(x, w) \subset z$: it is what makes $\mathbb{E}[g \mid x, w] = 0$, and that is what makes the moment insensitive to errors in h_s (next slide).
- ▶ Practical: $\hat{g}$ is two cross-fit regressions of $\Delta\eta$ (random forest, boosting, or a sieve), one on everything and one on the controls, differenced.

Orthogonality and Cross-Fitting in One Slide

Two nuisance functions, h_s and g , are estimated and plugged in. Why does their estimation error not contaminate the test? (Chernozhukov et al. 2018 double machine learning, applied to a model-selection statistic.)

- **Cost direction.** Perturb $h_s \rightarrow h_s + \delta h$: the moment moves by $-\mathbb{E}[\delta h(x, w) \cdot \mathbb{E}[g \mid x, w]] = 0$ by construction of g . First-order error in the cost function **cannot** enter. (This is Neyman orthogonality, delivered by the FWL subtraction.)
- **Instrument direction.** Perturb g : the moment moves by $\mathbb{E}[\mu_m \delta g]$, zero for every δg iff $\mu_m \equiv 0$. No null is needed: a model that satisfies the CMR has its moment pinned at zero however badly g is estimated, so with a true model on the menu $\hat{g}$ -error moves **power**, not size. In the both-wrong null ($\bar{m}_1 = -\bar{m}_2 \neq 0$) a first-order term survives along the level direction $\mu_{m_1} + \mu_{m_2}$: strong rejections are safe, **marginal** ones from a slow learner are not. (Conditional on the training folds $\hat{g}$ is the instrument, and the test is exactly sized for equal fit in that direction.)
- **Cross-fitting.** Split the **clusters** (whole markets, never products within a market) into K folds; evaluate each observation at nuisance functions fit on the other folds. This removes the own-observation bias that in-sample fitting puts into a **squared** moment ($\mathbb{E}[\tilde{Q}] = Q + \text{Var}(\bar{m}) + \text{overfit}$), without complexity restrictions on the learner.
- **Rate.** Remaining bias is a product of two nuisance errors, negligible once each converges faster than $n^{-1/4}$ (Bickel–Ritov). Random forests and boosting on a few hundred thousand rows clear this comfortably.

Inference for the Scalar Test

Aggregate the integrand to **cluster totals** $S_c^m = \sum_{i \in c} w_i \hat{\omega}_i^m \hat{g}_i$, the cluster being a union of whole markets (cereal: markets, $C \approx 1900$), so $\bar{m}_m = \frac{1}{n} \sum_c S_c^m$ and $\Delta \tilde{Q} = \bar{m}_1^2 - \bar{m}_2^2$.

- ▶ **Analytic (delta method).** $T_{RV} = \sqrt{n} \Delta \tilde{Q} / \hat{\sigma}$ with $\hat{\sigma}^2$ from the cluster totals and $\nabla f = (2\bar{m}_1, -2\bar{m}_2)$. Valid away from the boundary; as $\bar{m}_m \rightarrow 0$, $\hat{\sigma} \rightarrow 0$ and it fails.
- ▶ **Bootstrap 1, multiplier on the influence function.** Rademacher weights e_c on the linearized contribution: reproduces the analytic SE, same boundary problem (Kline–Santos 2012 score bootstrap).
- ▶ **Bootstrap 2, score-level wild.** Perturb each model's **moment before squaring**, $\tilde{m}_m^* = \bar{m}_m + \frac{1}{n} \sum_c e_c \tilde{S}_c^m$, then square. Keeps the cross term *and* the quadratic term, so it self-adapts: normal far from the boundary, fat-tailed near it, with no pretest. The variance it adds relative to the linear term scales as $1/T_m^2$, the per-model moment t -ratio.
- ▶ **Bootstrap 3, wild cluster bootstrap- t .** Studentize every draw; the few-cluster refinement (Cameron–Gelbach–Miller 2008).
- ▶ Because a squared mean is biased upward, Bootstraps 2–3 recenter: report $\Delta \tilde{Q} - (\widehat{V}_1 - \widehat{V}_2)$ against the centered bootstrap law.
- ▶ No refits: every bootstrap only reweights cached cluster totals. Markups and first stages are computed once.

Algorithm (BCS 1.0)

For a pair (m_1, m_2) ; the menu test repeats over all pairs.

1. **Split.** Partition the C clusters into K folds (3 in cereal); every observation inherits its cluster's fold. The same split is reused for every first stage.
2. **Cost residuals, per model.** For each m and fold k , fit $\hat{h}_s^{(-k)}(\mathbf{x}, \mathbf{w}) = \hat{\mathbb{E}}[MC^m \mid \mathbf{x}, \mathbf{w}]$ on the other folds and residualize on fold k ; fixed effects (product, chain, year) absorbed by a within-transform first. Computed **once** per model and reused in every pair.
3. **Instrument, per pair.** Cross-fit the two projections of $\Delta\eta$ and difference them:
$$\hat{g} = \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid \mathbf{x}, \mathbf{w}, z] - \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid \mathbf{x}, \mathbf{w}].$$
4. **Moments.** Cluster totals S_c^m , sample moments $\bar{m}_m$, criteria $\tilde{Q}_m = \bar{m}_m^2$, contrast $\Delta\tilde{Q}$.
5. **Inference.** T_{RV} : analytic cluster SE by default, Bootstrap 2 or 3 near the boundary, cached totals only.
6. **Menu.** Pairwise matrix of contrasts $\rightarrow$ model confidence set: the models never significantly beaten.

Two competing markups (markdowns) $(\eta_{jt}^{m_1}, \eta_{jt}^{m_2})$. Which fit data better? (both may be misspecified)

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \tau \eta_{jt}^{m_1} + (1 - \tau) \eta_{jt}^{m_2} + \omega_{jt} \text{ where } \mathbb{E}[\omega_{jt} | \mathbf{x}_{jt}, \mathbf{w}_{jt}, \mathbf{z}_{jt}] = 0$$

The only IV that matters is the **predicted markup difference**, net of the cost controls, which we get from two first stages

$$g(\mathbf{z}_{jt}) = \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} | \mathbf{x}_{jt}, \mathbf{w}_{jt}, \mathbf{z}_{jt}] - \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} | \mathbf{x}_{jt}, \mathbf{w}_{jt}]$$

As long as we have some **exclusion restrictions** $\mathbf{z}_{jt}$ that:

- **Relevance:** Predict markups **beyond** the cost shifters $(\mathbf{x}_{jt}, \mathbf{w}_{jt})$.
- **Exclusion:** Do **not** affect marginal cost.

Rest of paper: make $h(\cdot)$ and $g(\cdot)$ as flexible as possible (cross-fit), partial the controls out of g , and get a Vuong test whose inference survives the degeneracy boundary.

Two competing markups (markdowns) $(\eta_{jt}^{m_1}, \eta_{jt}^{m_2})$. Which fit data better? (both may be misspecified)

$$p_{jt} - \eta_{jt}^{m_1} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt} \text{ if } \tau = 1 (m_1)$$

$$p_{jt} - \eta_{jt}^{m_2} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt} \text{ if } \tau = 0 (m_2)$$

The only IV that matters is the **predicted markup difference**, net of the cost controls, which we get from two first stages

$$g(z_{jt}) = \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathbf{x}_{jt}, \mathbf{w}_{jt}, z_{jt}] - \mathbb{E}[\eta_{jt}^{m_1} - \eta_{jt}^{m_2} \mid \mathbf{x}_{jt}, \mathbf{w}_{jt}]$$

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The full conditional-moment test (BCS 2.0)

The Full Conditional-Moment Test in One Slide (BCS 2.0)

Use all of $\mu_m(z) = \mathbb{E}[\omega^m \mid z]$, not one direction: $Q(\eta^m) = \|\mu_m\|^2$, $H_0 : Q(\eta^{m_1}) = Q(\eta^{m_2})$. The scalar BCS moment, the DMSS 2SLS quadratic form, and this norm are one direction, K directions, and all directions of the same object.

$$\widehat{Q}(\eta^m) = \frac{1}{C(C-1)} \sum_{c \neq c'} (u_c^m)' u_{c'}^m, \quad u_c^m = \widehat{\Sigma}^{-1/2} \sum_{j \in c} \widehat{\omega}_j^m b(z_j), \quad b \text{ mean-centered.}$$

- **Cross-cluster U-statistic:** products of residuals from different markets are unbiased for $\|\mu_m\|^2$; the own-cluster terms carry the estimation noise that inflates any plug-in (the “positive shift”), so they are deleted. Centering b removes an $O(1)$ bias from the intercept of h_s (FWL).
- **Why bother:** a scalar criterion is degenerate whenever two wrong models agree in one direction; the norm is degenerate only if they induce the **same function** $\mu(z)$. Smaller degeneracy set, more power, one cross-fit regression per model, nothing refit per pair.
- **The price:** under a correct model the criterion sits at its minimum and is not normal, so inference has two regimes (next slide).

Inference in One Slide: Two Regimes, One Bootstrap

$$\Delta \widehat{Q} - (Q_{m_1} - Q_{m_2}) = \underbrace{\frac{1}{C} \sum_c \psi_c}_{\sqrt{C}\text{-normal if } \sigma_{\text{lin}}^2 > 0} + \underbrace{\frac{1}{C(C-1)} \sum_{c \neq c'} \Psi(c, c')}_{\text{rate-}C \text{ weighted } \chi^2 \text{ if } \mu_{m_1} \equiv \mu_{m_2}}$$

- **Non-degenerate:** models distinguishable through z ; cluster-robust RV is fine. **Degenerate:** observational equivalence (weak instruments, or near-identical markups as in cereal); the linear term vanishes, $C\widehat{Q} \rightarrow \sum_r v_r (\chi_{1,r}^2 - 1)$, mean zero and right-skewed, so a normal cutoff under-covers the upper tail. The standard error differs by $\sqrt{C}$ between regimes and the real comparisons sit near the boundary.
- **One multiplier bootstrap** on both terms, $\Delta \widehat{Q}^* - \Delta \widehat{Q} = \frac{1}{C} \sum_c e_c \widehat{\psi}_c + \frac{1}{C(C-1)} \sum_{c \neq c'} e_c e_{c'} \widehat{\Psi}(c, c')$, one Rademacher weight per cluster shared across terms and models. It nests the linear RV and the score-wild bootstrap and interpolates between them with no pretest and no switch; the p -value comes from the bootstrap law, not a normal table; nothing is refit.
- **Diagnostics:** $\widehat{F} = \max_m C\widehat{Q}(\eta^m)$ (is any model detectably off?) and $\widehat{\sigma}_{\text{lin}}^2$ (can this pair be ranked?), both descriptive. **Open:** uniform validity across the boundary (the Shi 2015 analogue), demand-estimation uncertainty, a Hansen model confidence set.

**What identifies conduct in the
data**

The Menu Is an Internalization Ladder

In cereal the candidate conducts differ in **how much of the cross-price structure** the firm prices on:



- ▶ **All** of them, single-product included, price above marginal cost. The test ranks models by the recapture they internalize, not by whether market power exists.
- ▶ So “single-product fits the moment best” is a statement about **cross-price internalization**, not a rejection of market power, and not a statement that firms ignore their sister brands in any literal sense.
- ▶ Perfect competition is on the menu as a straw man only (the constant-markup degeneracy). Cournot vs. Bertrand would be a different ladder.
- ▶ This is the substantive content of a criterion $\|\mathbb{E}[\omega^m \mid z]\|^2$ here: it orders the ladder; it does not estimate a markup level. The markup level comes from demand.

Near-Collinear Markups Make Instrument Strength Decisive

- ▶ Across the menu the implied markups are ≥ 0.92 correlated. The markup difference $\Delta\eta = \eta^{m_1} - \eta^{m_2}$ is small relative to η^m , so every comparison lives near the observational-equivalence boundary.
- ▶ Power requires **both** low $\text{corr}(\eta^{m_1}, \eta^{m_2})$ **and** $\Delta\eta \in \text{span}(z)$ net of the controls. The second is **instrument strength for conduct**, a different object from strength for demand: an instrument can move prices and shares beautifully and project onto $\Delta\eta$ not at all.
- ▶ Monte Carlo: rejection $1.00 \rightarrow 0.05$ as $\text{corr}(\eta^{m_1}, \eta^{m_2})$ rises from 0.97 to 0.999. That is the whole game.
- ▶ Practical implication: before testing, regress $\Delta\eta$ on the instruments (net of controls) and look at the fit. If nothing explains the markup difference beyond the cost shifters, no test, scalar or norm, will rank the pair. This is the DMSS weak-instruments-for-testing diagnostic in plain clothes.

Validity Is Orthogonality to the Cost Shock, Not to Price

A conduct instrument must be excluded from ω ; it **must** correlate with price, since a useful instrument moves markups. Choosing price-orthogonal instruments manufactures weak ones. The cereal instrument groups, after partialling (x , w , FE) out:

Group	What it is	Strength for conduct
opt_iv	demand-side Chamberlain optimal IV, $E[\partial\xi/\partial\theta \mid z]$ at a flexible $\hat{E}[p \mid z]$	strong ; the only group that isolates a singleton MCS
demog	chain-level income and demographic moments	strong; loses most of its variation to the FE
opc	price of a grain the product does <i>not</i> use (a rival's cost shifter)	valid and well motivated, but only $\approx 10\%$ of its variance survives the FE
diff	BLP / Gandhi-Houde differentiation instruments	weakest : degenerate for conduct

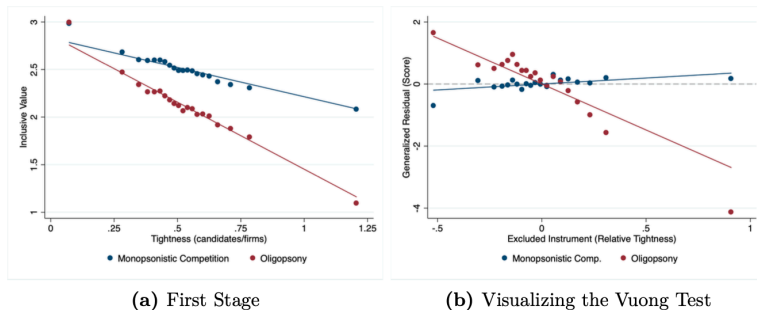
- ▶ The optimal IV is not circular: demand was estimated **without** a supply side, so $\hat{E}[p \mid z]$ is a reduced-form fit, not an equilibrium price under an assumed conduct. Its near-perfect span of expected prices is the source of its power.
- ▶ The BLP instruments are the demand workhorse and the conduct runt: their conduct content is nonlinear and collinear with the FE. Their apparent in-sample strength is squared-moment inflation.
- ▶ Partial the controls out of the instrument basis (FWL), as the scalar test partials $\hat{g}$; otherwise control overlap inflates the Gram matrix and groups are not comparable. Partialling **sharpens** valid strong instruments and exposes FE-collinear ones as weak.

Fixed Effects, Weighting, and Getting Out of Degeneracy

- ▶ With a low-dimensional h_s the cereal test is degenerate on every instrument: $\widehat{F} \leq 0$. Absorbing chain and year fixed effects in h_s strips confounding cost variation and lifts $\widehat{F}$ to +1.3; product fixed effects go further. A more flexible cost function can move a comparison from the degenerate into the non-degenerate regime, because it removes cost heterogeneity that was masquerading as moment violation.
- ▶ The same absorption removes each instrument's FE-collinear variation: the chain-level demographics and the product-level grain prices lose most of theirs, so fewer models can be ruled out with them. Only the optimal IV keeps enough product-level variation to isolate a single model.
- ▶ Servings weighting roughly halves the criterion magnitudes and leaves the ordering alone. Across a 20-cell sweep (5 FE sets $\times$ 2 weightings $\times$ 2 strong instruments) the model confidence set is {single} in 19 cells.
- ▶ Lesson: the **specification** of h_s (what is a cost control, which FE, how flexible) is as much a part of the identification argument as the instruments. Report it, every time.

Examples

Figure 5: Vuong Test



Note: Panel (a) plots the “first stage” relationship between the model-implied inclusive values Λ_i and Λ_i^{-j} and the instrumental variable t_{ij} , conditional on firm covariates z_j and candidate covariates x_i and two-week period dummies. Panel (b) plots the relationship between generalized residuals and the excluded instrument for the non-predictive monopsonistic competition and oligopsony models. Under proper specification, the correlation of the generalized residuals and the excluded instrument should be zero (the dashed line). The larger the deviation from zero, the greater the degree of mis-specification of the model.

- Offered wages for online job platform
- Compares Monopsony vs. oligopolistic competition vs. perfect comp.
- Compares tailored offers vs. not (price discrimination).

- Firms ignore competitors (Monopsony)
- Firms offer wages independent of candidate characteristics (experience, demographics).
- Firms are definitely NOT paying MPL.

Table 4: Non-Nested Model Comparison Tests ([Rivers and Vuong, 2002](#))

Model	(1) Monopsonistic Comp.	(2) Monopsonistic Comp.	(3) Oligopsony	(4) Oligopsony
	Not Predictive	Type Predictive	Not Predictive	Type Predictive
Perfect Competition	-54.84	-54.40	-39.92	-39.91
Monopsonistic, Not Predictive	–	7.83	3.98	2.69
Monopsonistic, Type Predictive		–	2.77	1.54
Oligopsony, Not Predictive			–	-3.67
Oligopsony, Type Predictive				–

Note: This table reports test statistics from the [Rivers and Vuong \(2002\)](#) non-nested model comparison procedure. Positive values imply the row model is preferred to the column model. Under the null of model equivalence, the test statistics are asymptotically normal with mean zero and unit variance.

specification. In the figure, the generalized residuals for the monopsonistic competition alternative are closely aligned with the x-axis, while the generalized residuals for the oligopsony alternative are strongly negatively related to tightness.

Our tests therefore suggest that models of firm behavior in which firms both ignore strategic interactions in wage setting and do not tailor wage offers to candidates on the basis of predictable preference variation are closer approximations to firms' true bidding behavior on the platform than are models in which firms act strategically and tailor offers. In [Appendix F](#), we report the testing results using the original [Vuong \(1989\)](#) likelihood comparison test, which yield qualitatively identical model comparisons. In the following analysis, we adopt the not-predictive monopsonistic competition model as our preferred model of conduct.

Do cartels encourage entry with high prices? Starc and Wollmann (2022) apply the BCS test to the generic-drug cartel; the menu is competitive vs. joint-profit-maximizing pricing before and after cartel formation, with non-members best responding.

*in 2013, Teva Pharmaceuticals, the largest generic firm, hired NP, a marketing executive with especially strong industry relationships, and tasked her with "price increase implementation."*¹ Over an 18-month period, industry participants exchanged thousands of calls and texts—alongside countless LinkedIn, Facebook, and WhatsApp messages and face-to-face conversations—with contacts at rival firms to coordinate the increases (Complaint, page 322).² Following this period, prescription drug expenditures by governments, private insurers, and individuals rose sharply by billions of dollars.

Starc Wollmann: Testing Results (prices rise with the cartel, less in markets that invite entry)

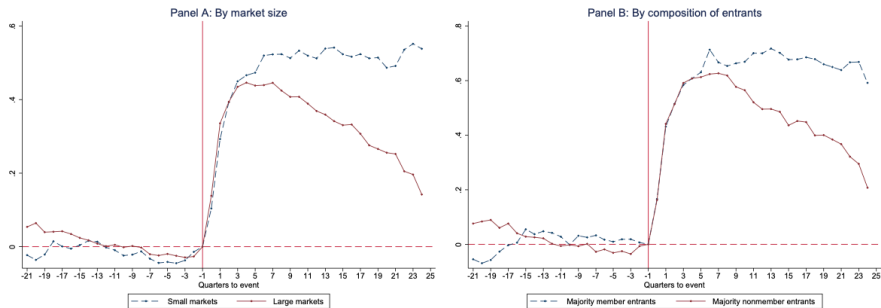


Figure III: Prices after cartel formation

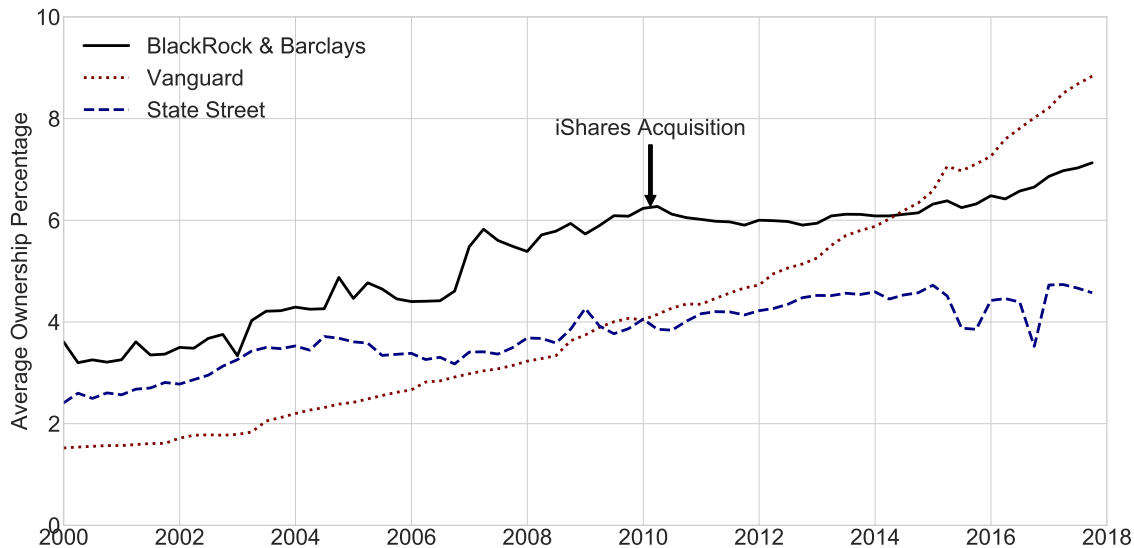
In this figure, we plot estimates of β^{τ} , which are derived from equation [1](#), against x^{τ} , which represents event time in quarters. For Panel A, we proxy for market size by computing the number of prescriptions filled in the period just prior to NP joining Teva, and we distinguish between large and small cartelized markets based on how this figure relates to the median value. For Panel B, we calculate the fraction of entrants that arrive as a result of ANDAs filed after NP joins Teva and identify markets in which a majority of those firms are members and nonmembers, respectively.

Application: Common Ownership (Backus Conlon Sinkinson)

What is the Common Ownership Hypothesis?

- ▶ Economics starts from the premise that firms maximize profits.
 - Friedman (1953): natural selection of firms and billiards players.
 - Or, firms answer to investors, **maximize shareholder value**.
 - We generally think of these objectives as aligned (possibly subject to managers' agency frictions).
- ▶ So what do investors want?
 - Some (diversified) investors may hold stakes in you *and your competitor*. These are called “common owners.”
 - Common owners are harmed when managers engage in **business stealing**.
- ▶ As a theory of firm behavior in **joint ventures** this is an old idea. The recent innovation is to extend this approach to **passive or institutional investors**

Rise of the Big Three (S&P 500 components, BCS AEJ:Micro 2021)



Some Reasonable (?) Assumptions (BCS 2020 P&P)

Most models of common ownership (implicitly) make the following assumptions:

Investor Portfolios Investors are entitled to a share of profits β_{fs} and hold portfolios $V_s = \sum_f \beta_{fs} \pi_f$

Manager Agency (Rotemberg 1984) Managers maximize a (γ) weighted average of investor payoffs

$$Q_f = \sum_s \gamma_{fs} V_s.$$

These are pretty general since s could contain the manager herself.

Derivation of Profit Weight

Easy to show:

$$\begin{aligned} Q_f(x_f, x_{-f}) &= \sum_{\forall s} \gamma_{fs} \cdot V_s(x_f, x_{-f}) \\ &\propto \pi_f + \sum_{g \neq f} \underbrace{\left(\frac{\sum_{\forall s} \gamma_{fs} \beta_{gs}}{\sum_{\forall s} \gamma_{fs} \beta_{fs}} \right)}_{\equiv \kappa_{fg}(\gamma_f, \beta)} \pi_g \end{aligned}$$

- ▶ κ_{fg} is interpreted as a **profit weight**, where one dollar of profits at firm g are valued as κ_{fg} dollars in the objective function of firm f .
- ▶ Depends on two primitives: β and γ .
 - β (cash-flow rights) is often directly observable.
 - γ embeds a model of corporate governance. Literature assumes $\gamma = \beta$ (“proportional control”).
- ▶ Not specific to a particular strategic game: this is the $\mathcal{H}(\kappa)$ from our multiproduct Bertrand FOCs.

The Reduced-Form Literature (and its Critics)

Symmetric homogeneous-good Cournot with profit weights gives a share-weighted average markup (with market demand elasticity ϵ):

$$\sum_f s_f \cdot \frac{p - mc_f}{p} = \frac{1}{\epsilon} \underbrace{\mathbf{s}' \mathcal{H}(\kappa) \mathbf{s}}_{\equiv MHHI}, \quad MHHI\Delta \equiv MHHI - \underbrace{\mathbf{s} \cdot \mathbf{s}}_{HHI}$$

Some regress

$$\log(p_{rjt}) = \beta_1 \cdot MHHI\Delta_{rt} + \beta_2 \cdot HHI_{rt} + \theta \cdot X_{rjt} + \alpha_t + \nu_{rj} + u_{rjt}$$

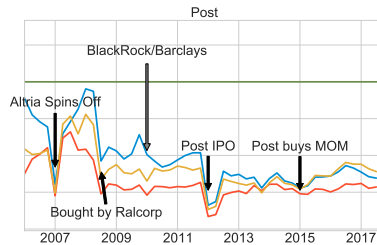
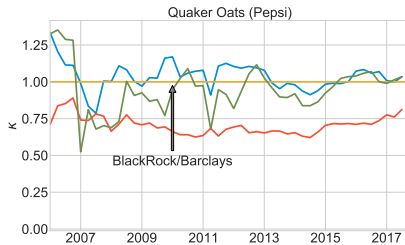
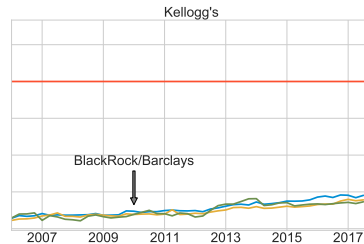
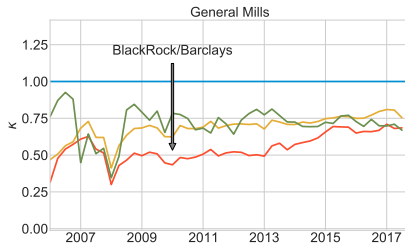
- ▶ Azar, Schmalz, Tecu (JF 2018): airline routes with more common ownership have 3–7% higher fares.
- ▶ But $MHHI\Delta$ is a function of **equilibrium market shares** – endogenous! (O'Brien Waehrer 2017). Price-concentration regressions all over again.
- ▶ Dennis, Gerardi, Schnitzler (JF 2022): not robust (driven by the share component); Kennedy, O'Brien, Song, Waehrer (2017): structural version finds nothing.
- ▶ Better idea: treat CO vs. own-profit maximization as a **conduct testing** problem.

Main Dataset is NielsenIQ (from Kilts) from 2007-2017

- ▶ Consolidate to dma-chain-week.
- ▶ Keep largest chains who price at chain level
 - Select based on # observations from panelist data.
- ▶ Consolidate upc → brand (Honey Nut Cheerios) from multiple package sizes and box designs.
 - Divide revenue by servings
 - Maintain the fiction that households purchase servings.

Why cereal? Long history as IO “model organism” (Nevo x 3), 4 big firms, and non-Nash behavior isn’t so crazy (FTC case in the 1970s, 1990s cartel/price war).

Cereal Data: Variation in κ



— General Mills — Kellogg's — Quaker Oats — Post

Implementation: Demand Specification

$$u_{ijt} = h_d(\mathbf{x}_{jt}^{(1)}, \mathbf{v}_{jt}; \theta_1) - \alpha p_{jt} + \lambda \log(\text{ad}_{jt}) + (\Sigma \nu_i + \Pi \mathbf{y}_i) \cdot \mathbf{x}_{jt}^{(2)} + \xi_{jt} + \varepsilon_{ijt}$$

- ▶ $\mathbf{y}_i$ demographics: estimated at the dma-chain-year level (from panelists); joint distribution of (*income*, *kids*).
- ▶ ν_i are random (normal) draws; price is lognormal.
- ▶ Lots of FE in $h_d(\cdot)$ (product, chain-dma, year, week of year); estimated with PyBLP plus 520 micro-moments for (Π, Σ) .
- ▶ Supply side/testing: exactly the machinery above – compute $\eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)$ under each conduct model once, then rank by the scalar RV test and by the full conditional-moment norm.

Possible Exclusion Restrictions

We are looking for variables which affect **demand but not supply**:

$$\sigma_j^{-1}(\mathbf{s}_t, \mathbf{p}_t, \mathbf{y}_t, \mathbf{x}_t, \mathbf{v}_t, \tilde{\theta}_2) = h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}; \theta_1) - \alpha p_{jt} + \lambda \log(\text{ad}_{jt}) + \xi_{jt}$$
$$p_{jt} - \eta_{jt}(\mathbf{s}_t, \mathbf{p}_t; \theta_2, \mathcal{H}_t(\kappa)) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}; \theta_3) + \omega_{jt}$$

Things we use (ranked by strength for conduct, see the instrument table):

- ▶ The demand-side optimal IV $\mathbb{E}[\partial \xi / \partial \theta \mid z]$ at a flexible $\hat{\mathbb{E}}[p \mid z]$: strongest, conduct-agnostic.
- ▶ Demographics (enter nonlinearly): $\mathbf{y}_t$ (chain-level income works well).
- ▶ Rivals' cost shifters $\mathbf{w}_{-j,t}$ (the price of a grain the product does not use): valid, thin after FE.
- ▶ Characteristics of other goods $f(\mathbf{x}_{-j,t})$ (BLP instruments): weakest for conduct.
- ▶ Obvious choice $\mathbf{v}_{jt}$ (product recalls): too few events to matter.

Things we don't use:

- ▶ Unobserved demand shocks ξ_{jt} (see MacKay Miller 2020 for $Cov(\xi_j, \omega_j) = 0$).
- ▶ Observable κ conduct shifters (financial mergers/events, see Miller Weinberg (2018))

Scalar Test Results (BCS 1.0): Rivers–Vuong $T, N(0, 1)$

	Others' Cost	Demographics	BLP Inst.	Dmd. Opt. Inst.
Own Profit Max vs.	Panel 1: $A(\mathbf{z}_t) = \mathbf{z}_t$, linear $h_s(\cdot)$			
Single Product	1.6489	1.1116	0.9977	1.6432
Common Ownership	-3.8928	-1.1957	0.5044	-1.0329
Double Marginalization	1.4435	0.9892	-0.0427	5.5684
Common Ownership + Double Marginalization	-0.1919	0.6815	0.1404	5.4688
Perfect Competition	1.1730	0.4171	0.7364	3.9589
Monopolist	-1.4097	-1.0680	-0.4523	-1.0908
Own Profit Max vs.	Panel 2: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12} \mathbf{z}_t]$, linear $h_s(\cdot)$ and $g(\cdot)$			
Single Product	1.4264	0.5795	0.6662	0.7516
Common Ownership	-2.3044	-0.5105	-0.0384	-1.4297
Double Marginalization	0.8644	0.4421	-0.5311	2.9800
Common Ownership + Double Marginalization	-0.9382	-0.2389	-0.3684	0.2460
Perfect Competition	0.7164	0.6135	-0.1080	1.8776
Monopolist	-0.8577	-0.4002	-0.3868	-1.1097
Own Profit Max vs.	Panel 3: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12} \mathbf{z}_t]$, random forest $h_s(\cdot)$ and $g(\cdot)$			
Single Product	5.0725	5.3347	5.2702	5.5758
Common Ownership	-3.9823	-3.6775	-4.0786	-4.3977
Double Marginalization	-6.3295	-9.9033	-6.5311	-7.6302
Common Ownership + Double Marginalization	-6.6568	-6.8735	-6.6624	-7.5578
Perfect Competition	-5.5005	-7.7749	-6.4083	-7.7653
Monopolist	-3.7240	-4.4602	-3.6134	-3.8959

Full Conditional-Moment Results (BCS 2.0): Per-Model Criterion and Model Confidence Set

Full-CMR criterion $C \cdot \widehat{Q}(\eta^m)$ (smaller = closer to $\mathbb{E}[\omega \mid z] = 0$) and the number of pairwise comparisons each model significantly loses (unified bootstrap); year+chain FE, boosting, controls partialled out of z , market clusters.

model	opt_iv $C\widehat{Q}$ (beaten)	demog (beaten)	opc (beaten)
single-product	34348 (0)	87 (0)	9.9 (0)
own-profit (Bertrand)	35473 (1)	106 (1)	12.4 (0)
common ownership κ	35590–35790 (1–4)	116–121 (1)	8.3–14.5 (0)
monopoly (big 4)	38261 (5)	138 (0)	12.7 (0)
monopoly (all)	50763 (6)	590 (6)	449 (6)
model confidence set	{single}	{single, monopoly-4}	all but monopoly-all

- **Own-profit is the robust substantive winner:** it beats common ownership, monopoly, and perfect competition on every instrument with power. Same ordering as the scalar test.
- Only the optimal IV has the power to isolate a singleton set; the weaker groups rank the top of the ladder but cannot separate it.
- Every $Q_m > 0$: this is “rank the misspecified models,” conditional on the demand-side instruments being excluded from the true cost shock.

The Single-Product Puzzle

- ▶ Single-product pricing (each brand ignores its sister brands) fits the moment condition **better** than own-profit maximization. Robust across instruments, samples, fixed effects, weighting, learners, and both criteria; look back at the scalar-test table.
- ▶ It is **not** “no market power”: single-product markups are $1/(\alpha(1 - \sigma_j))$ -sized, comfortably above cost. It says firms appear to internalize **less** cross-brand recapture than Bertrand with the estimated demand implies.
- ▶ Candidate explanations, none settled:
 1. Demand misspecification: the cross-price elasticities (recapture) are mismeasured, so the “correct” internalization looks wrong through the lens of this demand system. Recall the micro-data week: substitution patterns are the thing BLP gets least right.
 2. Rule-of-thumb pricing: brand managers price their own brand; the corporate parent does not reprice Cheerios when Lucky Charms moves.
 3. Another instrument or moment may reorder single vs. own-profit; nothing we have tried does.
- ▶ Methodological point: a result like this only becomes visible once the test is honest about ranking misspecified models. A specification test would have rejected everything and said nothing.

An Internalization Parameter

Let κ represent the weight a firm places on competitors and τ the internalization of those weights.

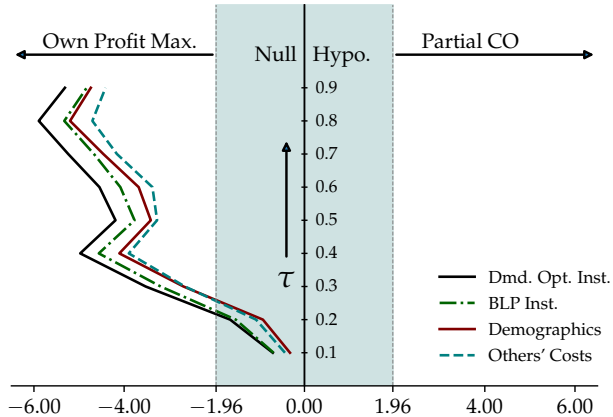
$$\arg \max_{p_j : j \in \mathcal{J}_f} \sum_{j \in \mathcal{J}_f} (p_j - mc_j) \cdot \sigma_j(\mathbf{p}) + \sum_{g \neq f} \tau \kappa_{fg} \sum_{j \in \mathcal{J}_g} (p_k - mc_k) \cdot \sigma_k(\mathbf{p})$$

Now,

- ▶ $\tau = 0$ implies own-profit maximization
- ▶ $\tau = 1$ implies common ownership pricing
- ▶ τ in between is...? Agency?

We test $\tau \in (0.1, \dots, 0.9)$ against own-profit maximization.

Internalization Parameter Results



Stepping Back: What About the Mechanism?

- ▶ In RTE cereal, we see strong evidence in favor of own-profit maximization rather than common ownership pricing, on BCS 1.0 and BCS 2.0 alike.
- ▶ In some sense this is a **frictionless** model
 - Managers do exactly what investment managers want
 - Investment managers seek to maximize **portfolio value**
- ▶ Some attempts at micro foundations are questionable
 - Anton, Ederer, Gine, Schmalz: toy model has manager choose effort to reduce mc .
 - Weaker managerial incentives: higher mc and higher p
 - Some correlation that executives at commonly owned firms have performance less correlated with profitability.
- ▶ In some sense CO is what we would see absent agency problems, so where are they?

Thanks!
